Statistics – 4

Quartiles, IQR, Outlier Detection, Skewness & Kurtosis

05 April 2026 · Lecture Notes · Statistics-4

Quartiles & IQR	Outlier Detection	Skewness	Kurtosis

Topic	Key Concept
Quartiles & IQR	Divide data into 4 parts; measure spread of middle 50%
Outlier Detection	Fence method: Q1–1.5×IQR and Q3+1.5×IQR
Skewness	Is data balanced or tilted to one side?
Kurtosis	Are extreme values common? (tail weight)

§ 01 Quartiles & IQR

QUARTILES

Q1 = 25th percentile — 25% of data lies below this value

Q2 = 50th percentile — median (middle value)

Q3 = 75th percentile — 75% of data lies below this value

Quartiles divide sorted data into **4 equal parts**.

IQR — INTERQUARTILE RANGE

IQR = Q3 – Q1

IQR = width of the region where the **majority of data lies**.

SD/Mean are affected by outliers. IQR is **robust** — not affected by extreme values.

The box in a boxplot = IQR region (between Q1 and Q3).

How to Calculate Quartiles — Formula

```
L = percentile_fraction × (n – 1) + 1
Q1: L = 0.25 × (n–1) + 1 Q3: L = 0.75 × (n–1) + 1
If L is fractional: interpolate between adjacent values
```

Worked Example — Calculating Q1, Q3, IQR

`Data (sorted, n=15): 4, 8, 12, 15, 18, 21, 24, 27, 30, 33, 38, 42, 47, 51, 94`

Quartile	L formula	L value	Interpolation	Result
Q1 (25th %ile)	0.25 × (15–1) + 1	4.5	Position 4=15, 5=18 → 15 + 0.5×(18–15) = 15 + 1.5	Q1 = 16.5
Q2 (50th %ile)	0.50 × (15–1) + 1	8	Position 8 = 27	Q2 = 27
Q3 (75th %ile)	0.75 × (15–1) + 1	11.5	Position 11=38, 12=42 → 38 + 0.5×(42–38) = 38 + 2	Q3 = 40
IQR	Q3 – Q1	40 – 16.5	—	IQR = 23.5

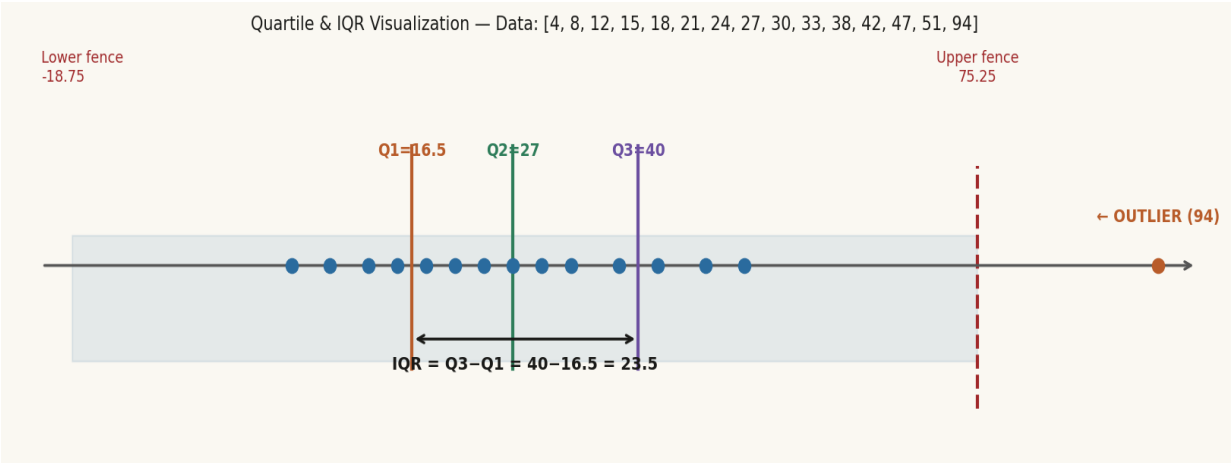

Figure 1: Quartile positions and IQR region — 94 is clearly beyond upper fence

§ 02 The Fence Method — Outlier Detection

FENCE FORMULAS

Lower Fence = Q1 – 1.5 × IQR

Upper Fence = Q3 + 1.5 × IQR

Any value below the lower fence or above the upper fence = **outlier / unusual data**.

Normal distribution: IQR×1.5 fences capture **99.3% of all data**. Outside = unusual.

Tukey (person) → 1.5 is the standard multiplier used worldwide.

Worked Example — Fences for the Dataset

Fence	Formula	Calculation	Value
Lower fence	Q1 – 1.5 × IQR	16.5 – 1.5 × 23.5 = 16.5 – 35.25	= –18.75
Upper fence	Q3 + 1.5 × IQR	40 + 1.5 × 23.5 = 40 + 35.25	= 75.25
Outlier check	Data: 94	94 > 75.25 → outside upper fence	94 = OUTLIER

```
Lower fence = –18.75 | Upper fence = 75.25 | Outlier: 94
```

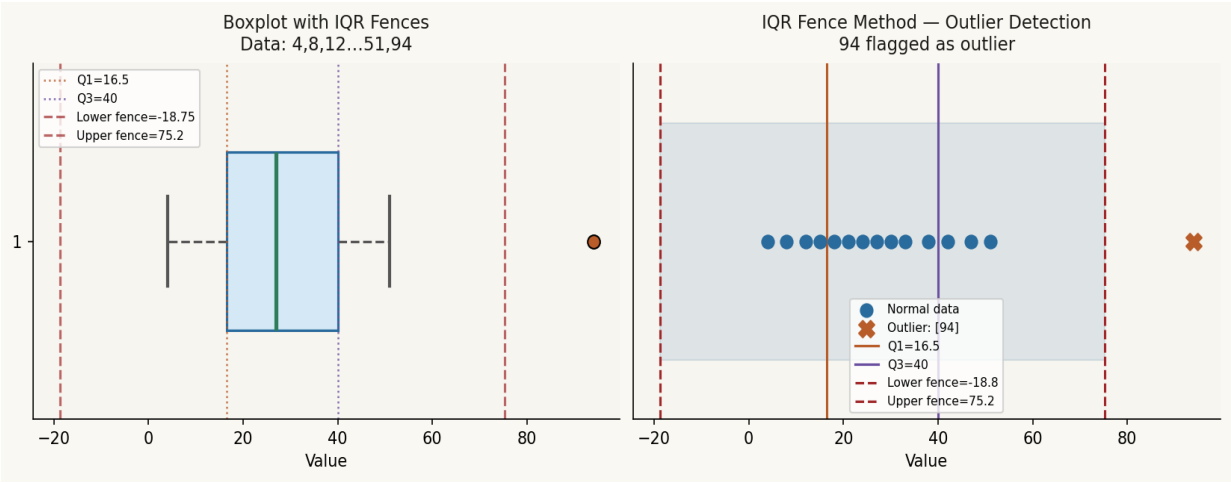

Figure 2: Boxplot with IQR fences and outlier detection (94 flagged)

Removing Outliers in Python (Boolean Masking)

PYTHON IMPLEMENTATION

arr = [1, 2, 3, 4, 5, 6] lower = 2 upper = 5

mask = (arr >= lower) & (arr <= upper)

mask = [F, T, T, T, T, F] → arr[mask] = [2,3,4,5] ← **clean data**

~mask = [T, F, F, F, F, T] → arr[~mask] = [1,6] ← **outliers extracted**

This is the standard pandas/numpy pattern for outlier removal.

Why Remove Outliers?

ML CONTEXT — WHY UNUSUAL DATA MATTERS

ML models learn patterns. Outlier data creates confusion and **biases the model**.

Example: Stroke prediction dataset (2000 patients) — Age, Veg/Non-veg, Active → Stroke %

If patient has age=300 (data entry error) → ML learns wrong pattern.

Debate: Removing outliers may remove critical rare cases (e.g., rare disease presentations).

→ T-shirt sizing insight: 1000 samples → 950 great + few unusual → clean them for ML.

→ Solution: *understand your data domain* before blindly removing outliers.

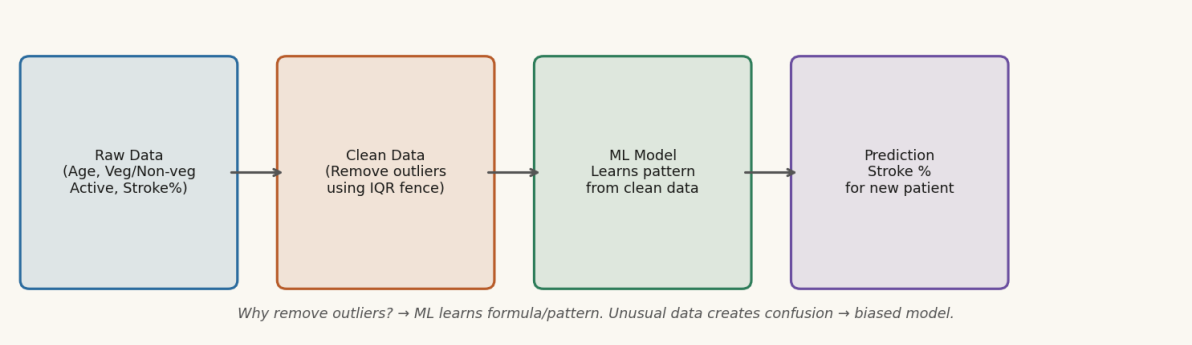

Figure 3: ML pipeline — why clean outliers before model training

§ 03 Skewness

DEFINITION

Skewness = is data balanced or tilted to one side?

Mean → center of data. SD → spread of data. Skewness → shape of data.

It tells us where the **tail** of the distribution points.

$$\text{Skewness} = \Sigma\,(x_i - \mu)^3 / (n \cdot \sigma^3)$$

x_i = individual data point | μ = mean | n = count | σ = std deviation

Cube (^{3}) preserves direction (sign) of deviation

Three Cases of Skewness

Case	Data Example	Mean	Median	Skew Value	Shape	Tail
Balanced (Symmetric)	40,50,60,70,80	60	60	≈ 0	Bell curve	Both equal
Right Skew (Positive)	10,20,30,90,90	38	30	> 0	Right tail	Mean > Median
Left Skew (Negative)	10,60,70,80,90	62	70	< 0	Left tail	Mean < Median

KEY RULE FOR SKEWNESS

Positive skew value → **Right skew** (tail extends right, few high outliers pull mean up)

Negative skew value → **Left skew** (tail extends left, few low outliers pull mean down)

Value ≈ 0 → **Symmetric data** (Normal distribution — mean = median)

Right skew: Mean > Median (outliers inflate mean) | Left skew: Mean < Median

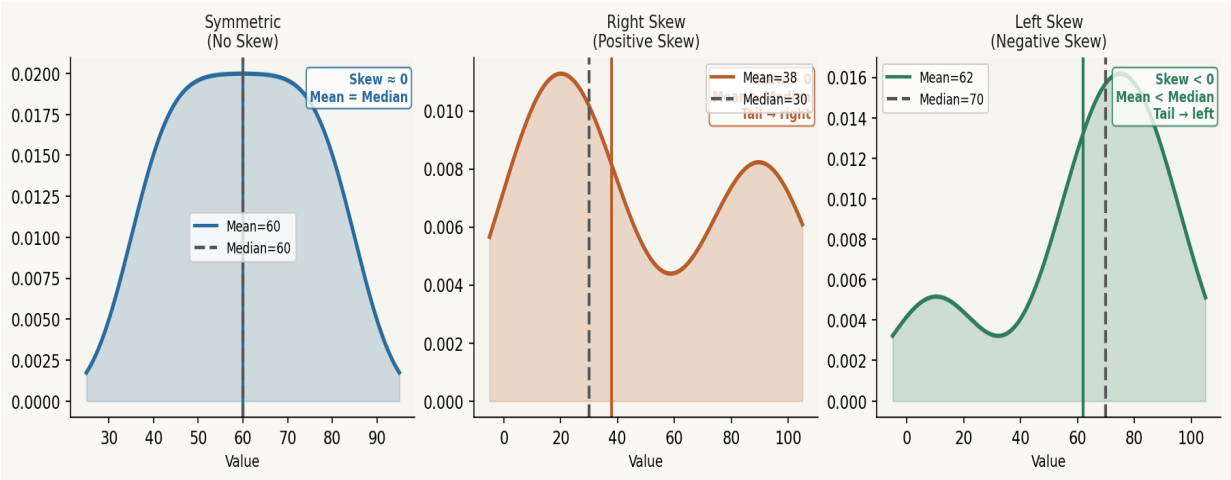

Figure 4: Three skewness cases — symmetric, right skew, left skew with mean/median positions

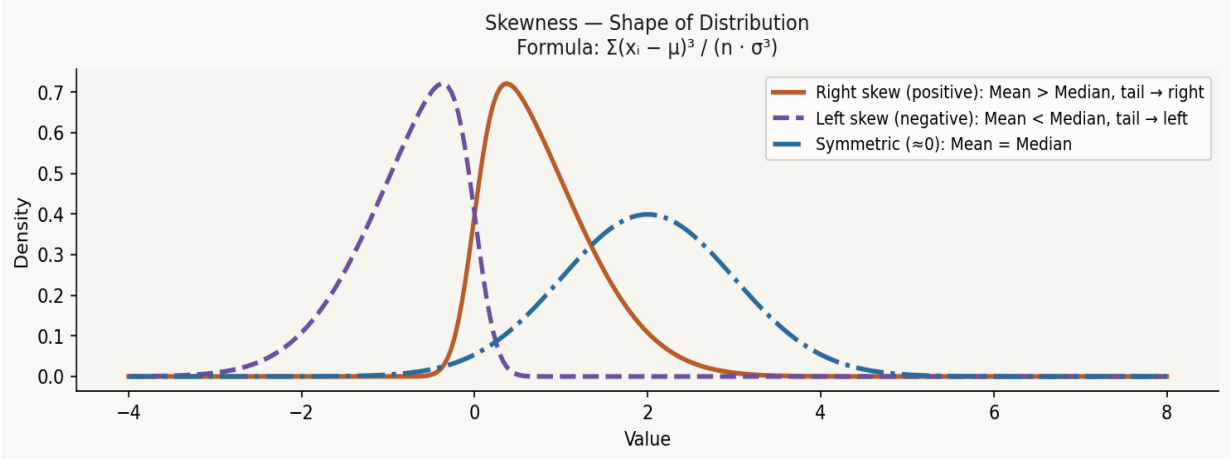

Figure 5: Skewness distributions — positive, negative, and symmetric overlaid

Understanding Skewness from a Histogram:

For a class of 100 students with Normal distribution: Poor marks → 25%, Moderate marks → 50%, Good marks → 25%.

The marks histogram would show a symmetric bell curve with equal tails on both sides.

§ 04 Kurtosis

DEFINITION

Kurtosis answers: *"Are extreme values common?"*

It measures the **weight of the tails** — how peaked or flat the distribution is.

Note: Kurtosis is not that reliable / hard to interpret in isolation.

```
Kurtosis = Σ (x■ − μ)■ / (n · σ■)
Power of 4 (■) removes sign → measures magnitude of tail extremity
Squaring (x²) removes directional sign from the value
```

Kurtosis Value	Interpretation	Distribution Shape
≈ 0 (Mesokurtic)	Normal — balanced tails	Standard bell curve
> 0 (Leptokurtic)	Few BIG outliers OR few small outliers — heavy tails	Peaked center, fat tails
< 0 (Platykurtic)	Flat distribution — thin tails, few outliers	Flat / wide shape

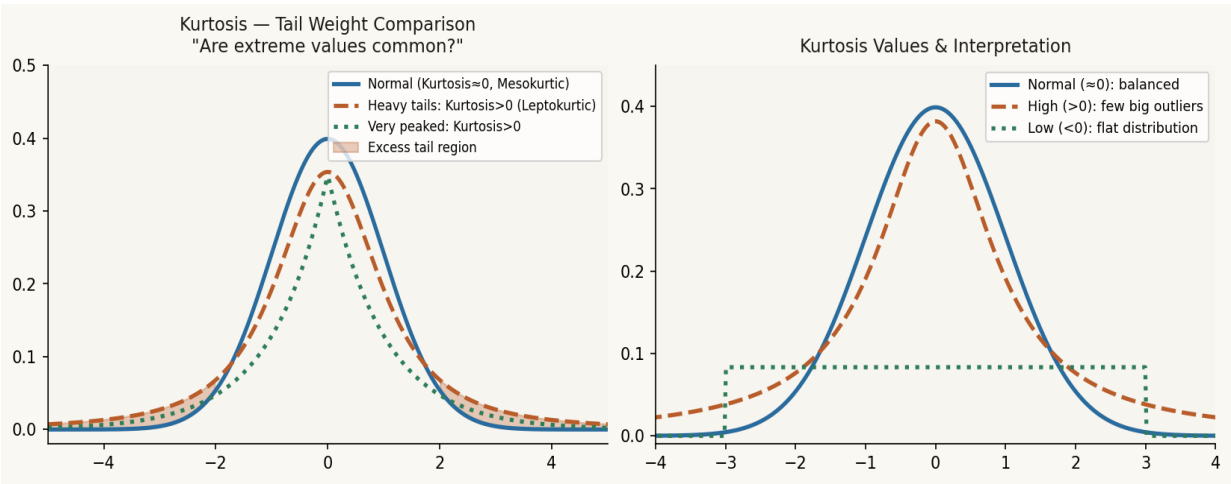

Figure 6: Kurtosis — heavy-tailed vs normal vs flat distributions

§ 05 Summary — Key Formulas & Concepts

Concept	Formula	Interpretation
Q1	25th percentile: L = 0.25(n–1)+1	25% data below
Q3	75th percentile: L = 0.75(n–1)+1	75% data below
IQR	Q3 – Q1	Middle 50% spread; robust to outliers
Lower Fence	Q1 – 1.5 × IQR	Anything below = outlier
Upper Fence	Q3 + 1.5 × IQR	Anything above = outlier
Skewness	$\Sigma(x_i-\mu)^3 / (n\cdot\sigma^3)$	>0: right skew | <0: left skew | ≈0: symmetric
Kurtosis	$\Sigma(x_i-\mu)^4 / (n\cdot\sigma^4)$	>0: heavy tails | <0: flat | ≈0: normal

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